

ORDYNSKOYE, Russian Federation

WMO: 297260

Lat: 54.3653N

Lon: 81.8842E

Elev: 435

StdP: 14.47

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-28.8	-40.3	0.6	-34.7	-34.0	0.8	-28.6	16.8	15.3	15.4	15.9	0.7	60	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	85.5	65.5	82.7	64.7	79.8	63.4	68.8	79.5	67.2	77.9	65.7	76.3	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	94.4	73.4	63.4	88.7	72.1	61.7	83.4	70.8	33.3	79.6	32.0	77.8	30.9	76.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.5	12.9	11.5	DB	-39.9	91.4	6.9	3.8	-44.9	94.1	-48.9	96.4	-52.8	98.5	-57.8	101.3
(5)				WB	-40.0	70.9	6.4	2.0	-44.6	72.3	-48.4	73.5	-52.0	74.6	-56.7	76.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	-0.7	4.9	20.2	39.1	53.4	64.0	67.1	62.5	51.0	38.2	20.4	6.7
	DBStd	26.06	16.15	14.06	12.17	9.73	9.47	7.54	5.73	6.52	7.95	8.83	14.24	15.29
	HDD50	6879	1573	1264	924	348	77	4	0	2	83	375	888	1341
	HDD65	10937	2038	1684	1389	779	371	107	44	126	425	830	1338	1806
	CDD50	1666	0	0	0	19	183	424	530	388	112	10	0	0
	CDD65	250	0	0	0	0	12	76	110	48	4	0	0	0
	CDH74	2465	0	0	0	9	244	753	875	505	79	0	0	0
	CDH80	594	0	0	0	1	58	181	212	125	17	0	0	0
Wind	WSAvg	4.9	4.7	4.7	5.2	5.3	5.3	4.5	4.0	4.1	4.4	5.1	5.7	5.5
Precipitation	PrecAvg	17.0	0.8	0.7	0.7	0.9	1.4	2.0	2.9	1.9	1.6	1.4	1.3	1.2
	PrecMax	23.1	1.5	1.3	1.4	2.0	3.2	3.7	5.7	4.8	3.8	3.3	2.7	2.5
	PrecMin	11.9	0.2	0.0	0.2	0.1	0.2	0.6	0.8	0.2	0.4	0.5	0.7	0.4
	PrecStd	3.1	0.4	0.3	0.4	0.5	0.8	1.0	1.3	1.1	0.8	0.7	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	35.7	49.3	74.3	86.2	87.9	89.9	89.2	82.3	67.5	49.2	35.6
		MCWB	32.3	32.1	40.2	53.6	60.8	65.2	67.5	67.2	61.1	52.7	43.7	33.2
	2%	DB	29.9	32.0	43.2	66.9	80.1	84.7	85.0	83.7	75.0	61.3	44.8	32.8
		MCWB	27.8	29.6	36.9	51.1	58.3	64.2	67.3	65.8	58.9	49.6	40.2	31.1
	5%	DB	26.6	29.2	39.4	60.9	75.9	81.7	81.9	79.7	69.6	56.2	40.4	30.1
		MCWB	25.0	27.2	34.6	48.7	56.8	63.8	66.2	64.0	56.3	47.3	37.3	28.5
	10%	DB	22.9	25.6	36.1	55.6	71.1	78.6	79.0	76.0	65.3	51.1	36.8	26.8
		MCWB	21.3	23.7	32.7	45.7	55.0	62.8	65.1	62.4	54.4	44.9	34.1	25.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	33.2	41.2	56.2	63.4	68.4	71.6	70.0	62.7	54.0	44.8	33.6
		MCDB	34.2	34.8	48.6	70.0	80.3	80.8	82.8	81.5	74.6	64.5	47.9	34.6
	2%	WB	28.4	29.8	37.4	52.3	60.7	66.9	69.6	67.5	59.6	50.7	40.3	31.5
		MCDB	29.7	31.9	41.7	63.8	74.9	79.2	79.8	79.7	72.3	59.4	43.8	32.8
	5%	WB	25.0	27.2	35.1	49.1	58.3	65.5	68.1	65.5	57.4	48.1	37.6	28.5
		MCDB	26.7	29.1	38.8	60.0	72.3	77.7	77.8	76.8	68.1	54.9	40.6	29.8
	10%	WB	21.3	23.8	32.9	46.0	56.2	64.1	66.6	63.7	55.1	45.4	34.3	25.2
		MCDB	22.9	25.6	36.0	55.2	69.4	75.5	76.3	74.1	63.4	51.1	36.7	26.7
Mean Daily Temperature Range	5% DB	MDBR	16.9	19.1	19.2	18.9	21.8	19.2	17.6	19.4	18.6	14.8	11.9	14.4
		MCDBR	14.7	16.1	20.3	29.2	29.9	25.0	22.2	25.9	27.8	25.5	12.7	13.9
		MCWBR	13.9	14.5	15.0	16.1	12.8	9.6	8.3	10.9	13.6	15.1	9.6	13.3
	5% WB	MCDBR	14.7	15.4	17.7	27.1	25.0	21.0	18.1	21.8	24.2	22.9	12.0	13.0
		MCWBR	14.0	14.2	13.5	15.7	11.9	8.8	8.0	9.9	12.5	14.4	9.6	12.9
Clear-Sky Solar Irradiance	taub	0.296	0.329	0.380	0.410	0.410	0.411	0.437	0.413	0.372	0.355	0.331	0.298	
	taud	2.011	1.961	1.904	2.097	2.203	2.287	2.213	2.255	2.317	2.235	2.053	1.960	
	Ebn at Noon	191	227	242	257	265	266	255	254	250	221	174	157	
	Edh at Noon	23	35	47	44	42	39	42	37	31	27	23	20	
All-Sky Solar Radiation	RadAvg	319	653	1116	1493	1749	1935	1737	1482	1019	517	253	210	
	RadStd	27	38	85	89	187	163	111	104	112	72	43	31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page